Nolan Dey

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Objective

Focused on maximizing business impact above replacement through $0 \rightarrow 1$ projects that are high-effort and high-value. Currently working at the intersection of ML research and hardware performance modelling to enable effective hardware-software codesign through novel LLM training and inference methods.

Full-time experience

Staff ML Research Scientist — Cerebras Systems (Toronto, Canada) Mar 2025 - Present

- Leading research into training and accelerating text diffusion models on Cerebras hardware, boosting inference profitability and generation speeds
- First author of CompleteP [7] and second author of Power lines [5] which together enabled 5x compute effiency improvements for 200B-375B customer training runs

Senior ML Research Scientist — Cerebras Systems (Toronto, Canada) Oct 2023 - Mar 2025

- Led research into weight sparse and activation sparse training to establish a competitive advantage over GPUs using Cerebras' unique acceleration of unstructured sparse matrix multiplications
- Identified major confounder (systematic hyperparameter detuning) in nearly every past weight sparse training work, then proposed SμPar [10] to solve this issue and enable more robust weight sparse training research

ML Research Scientist II — Cerebras Systems (Toronto, Canada) Sep 2022 - Oct 2023

- First author of CerebrasGPT [12], demonstrating the first scalable LLM training on Cerebras hardware and leading to $900M+ deal with G42 in revenue (85% of total revenue at the time)
- Developed training recipe for BTLM-3B [13], an LLM competitive with contemporary 7B LLMs with 2.5x more parameters, 3.3x more training compute, and 1.6x more training tokens. This training recipe laid the foundation for compute efficient customer pretraining runs at 13B-70B scale

ML Research Scientist I — Cerebras Systems (Toronto, Canada) Sep 2021 - Sep 2022

- Developed novel dynamic weight-sparse training algorithms uniquely accelerated by Cerebras hardware

Internship experience

Teaching Assistant — University of Waterloo (Waterloo, Canada) Jan 2020 - Sep 2021

- Full TA for SYDE 462 (Systems Design Capstone Project 2), full TA for SYDE 461 (Systems Design Capstone Project 1), full TA for SYDE 361 (Engineering Design), and half TA for SYDE 223 (Algorithms and Data Structures)

Machine Learning Research Intern — Mind Foundry (Greater Oxford Area) Jun 2018 - Aug 2018

- Augmented AutoML system with method to predict the training time of ML algorithms

Machine Learning & Web Intern — Apple (Cupertino, California) Sep 2017 - Dec 2017

- Identified and automated a costly labelling process with image classifier deployed on MacOS
- Created a scheduling application with Ember, Rails, and D3 which improved internal workflows

Data Science Intern — Capital One (Kitchener, Canada) Jan 2017 - Apr 2017

- Developed and deployed a distributed event data pipeline that processed 480 MB/s and saved $200k/year

Full-Stack Developer Intern — Parabol (Remote) Sep 2016 - Dec 2016

- Feature development for open-source web-app using Node, React, Redux, RethinkDB, GraphQL

Software Engineering Intern — Connected Lab (Toronto, Ontario) May 2016 - Aug 2016

- Developed web-apps using Node, React, MongoDB and an iOS app for streaming music to multiple speakers

iOS Developer Intern — Kik Interactive, Inc. (Waterloo, Ontario) Sep 2015 - Dec 2015

- Wrote major social sharing feature in Objective-C for an iOS app with 200M users

Quality Assurance Engineering Intern — Kik Interactive, Inc. (Waterloo, Ontario) Jan 2015 - Apr 2015

- Rigorously tested new features, conducted usability testing, and worked with developers to resolve issues

Summer Engineering Intern — Hatch (Mississauga, ON) Aug 2013

- Performed internal and external literature review to document corrosion prevention solutions in smelting furnaces

Education

MASc - Systems Design Engineering — University of Waterloo (Waterloo, Canada) Sep 2019 - Sep 2021

- GPA: 4.0/4.0
- Topic: Studying CNN representations through activation dimensionality reduction and visualization
- Courses: Simulating Neurobiological Systems, Neural Networks, Advanced Image Processing, Time Series Modelling

BASc - Systems Design Engineering — University of Waterloo (Waterloo, Canada) Sep 2014 - Apr 2019

- GPA: 3.7/4.0
- Exchange semester at Lund University (Jan 2018 - May 2018)
- Courses: Machine Intelligence, Applied Artificial Intelligence, Image Processing, Optimization

Publications

[1] Shane Bergsma, **Nolan Dey**, Joel Hestness. "Predicting Training Re-evaluation Curves Enables Effective Data Curriculums for LLMs". In: *ICLR*. 2026. URL: https://openreview.net/forum?id=wjaTz8nYjD.

[2] Shane Bergsma, Bin Claire Zhang, **Nolan Dey**, Shaheer Muhammad, Gurpreet Gosal, Joel Hestness. "Scaling with Collapse: Efficient and Predictable Training of LLM Families". In: *ICLR*. 2026. URL: https://openreview.net/forum?id=3YKeB9R1g9.

[3] Mostafa Elhoushi, Alexander D. Pretko, **Nolan Simran Dey**, Bin Claire Zhang, Gavia Gray, Gurpreet Gosal, Abdulrahman Mahmoud, Shane Bergsma, Joel Hestness. "Don't Drop Dropout: Optimizing Layer Sparsity for Efficient LLM Training and Inference". In: *ICML*. 2026. URL: https://openreview.net/forum?id=AEHn9ya4Re.

[4] Subham Sekhar Sahoo, Lingjie Chen, Khiem Pham, Jonathan Geuter, Chaitanya Dwivedi, Varad Pimpalkhute, Yash Akhauri, Alexander Moreno, Mikhail Yurochkin, Zhenting Wang, Mostafa Elhoushi, **Nolan Dey**, Shane Bergsma, Joel Hestness, John Thickstun, Eric Xing, Zhengzhong Liu. *Unlocking Lossless Speedups in LLMs via Discrete Diffusion*. 2026. arXiv: 2609.04010 [cs.LG]. URL: https://arxiv.org/abs/2609.04010.

[5] Shane Bergsma, **Nolan Dey**, Gurpreet Gosal, Gavia Gray, Daria Soboleva, Joel Hestness. "Power Lines: Scaling Laws for Weight Decay and Batch Size in LLM Pre-training". In: *NeurIPS*. 2025. URL: https://arxiv.org/abs/2505.13738.

[6] Shane Bergsma, **Nolan Simran Dey**, Gurpreet Gosal, Gavia Gray, Daria Soboleva, Joel Hestness. "Straight to Zero: Why Linearly Decaying the Learning Rate to Zero Works Best for LLMs". In: *ICLR*. 2025. URL: https://openreview.net/forum?id=hrOlBgHsMI.

[7] **Nolan Dey**, Bin Claire Zhang, Lorenzo Noci, Mufan Li, Blake Bordelon, Shane Bergsma, Cengiz Pehlevan, Boris Hanin, Joel Hestness. "Don't be lazy: CompleteP enables compute-efficient deep transformers". In: *NeurIPS*. 2025. URL: https://arxiv.org/abs/2505.01618.

[8] **Nolan Simran Dey**, Eric Taylor, Alexander Wong, Bryan P. Tripp, Graham W. Taylor. "Neuron-based explanations of neural networks sacrifice completeness and interpretability". In: *TMLR* (2025). ISSN: 2835-8856. URL: https://openreview.net/forum?id=UWNa9Pv6qA.

[9] **Nolan Dey**, Quentin Anthony, Joel Hestness. *The practitioner's guide to the maximal update parameterization*. https://www.cerebras.ai/blog/the-practitioners-guide-to-the-maximal-update-parameterization. 2024.

[10] **Nolan Dey**, Shane Bergsma, Joel Hestness. "Sparse maximal update parameterization: A holistic approach to sparse training dynamics". In: *NeurIPS*. 2024. URL: https://arxiv.org/abs/2405.15743.

[11] Esha Singh, Shane Bergsma, **Nolan Simran Dey**, Joel Hestness, Gavia Gray. "Empirical Upper Bounds for Unstructured Sparsity in Compute-Efficient Language Modeling". In: *Workshop on Machine Learning and Compression, NeurIPS 2024*. 2024. URL: https://openreview.net/forum?id=qOnKSqiGtR.

[12] **Nolan Dey**, Gurpreet Gosal, Zhiming, Chen, Hemant Khachane, William Marshall, Ribhu Pathria, Marvin Tom, Joel Hestness. *Cerebras-GPT: Open Compute-Optimal Language Models Trained on the Cerebras Wafer-Scale Cluster*. 2023. arXiv: 2304.03208 [cs.LG]. URL: https://arxiv.org/abs/2304.03208.

[13] **Nolan Dey**, Daria Soboleva, Faisal Al-Khateeb, Bowen Yang, Ribhu Pathria, Hemant Khachane, Shaheer Muhammad, Zhiming, Chen, Robert Myers, Jacob Robert Steeves, Natalia Vassilieva, Marvin Tom, Joel Hestness. *BTLM-3B-8K: 7B Parameter Performance in a 3B Parameter Model*. Efficient Natural Language and Speech Processing NeurIPS Workshop. 2023. arXiv: 2309.11568 [cs.AI]. URL: https://arxiv.org/abs/2309.11568.

[14] Faisal Al-Khateeb, **Nolan Dey**, Daria Soboleva, Joel Hestness. *Position Interpolation Improves ALiBi Extrapolation*. 2023. arXiv: 2310.13017 [cs.CL]. URL: https://arxiv.org/abs/2310.13017.

[15] Daria Soboleva, Faisal Al-Khateeb, Robert Myers, Jacob R Steeves, Joel Hestness, **Nolan Dey**. *SlimPajama: A 627B token cleaned and deduplicated version of RedPajama*. https://www.cerebras.net/blog/slimpajama-a-627b-token-cleaned-and-deduplicated-version-of-redpajama. 2023. URL: https://huggingface.co/datasets/cerebras/SlimPajama-627B.

[16] **Nolan Dey**, J. Eric Taylor, Bryan P. Tripp, Alexander Wong, Graham W. Taylor. *Identifying and Interpreting Tuning Dimensions in Deep Networks*. Shared Visual Representations in Human & Machine Intelligence Workshop, NeurIPS. 2020. URL: https://arxiv.org/abs/2011.03043v2.

[17] Victor Reyes Osorio, Rajan Iyengar, Xueyang Yao, Presish Bhattachan, Adrian Ragobar, **Nolan Dey**, Bryan Tripp. "37,000 Human-Planned Robotic Grasps With Six Degrees of Freedom". In: *IEEE Robotics and Automation Letters* 5.2 (2020), pp. 3346–3351. DOI: 10.1109/LRA.2020.2976295.

Skills

- **Research:** Python, PyTorch, Jax, TensorFlow, Keras, SKLearn, MatPlotLib, NumPy, SciPy, Pandas, MATLAB
- **Infrastructure:** SLURM, GCP, AWS, Docker, Kafka, Terraform, PostgreSQL, Git, Linux, GraphQL, MongoDB
- **Software Engineering:** Node, React, Redux, Javascript, multiprocessing, Objective-C, Ember, D3, Rails, Scala [5]

Volunteer Experience

- **Organizer** *May 2026 - Dec 2026*
  BeNTo: Beyond next token prediction - Diffusion and flow models for next-generation decoding, NeurIPS 2026 workshop
- **Reviewer** *May 2025 - Aug 2025*
  NeurIPS 2025
- **Founder & Head Organizer** *Dec 2012 - Dec 2019*
  Carols for Cans
  - Annually organized event where students sing Christmas carols and ask for food donations
  - Donated over 10,314 food items to GTA food banks since 2012, with 599 students participating over 8 years!